

Date: _____

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$$m_x(t) = \left(\sum_{i=1}^K p_i e^{t_i} \right)^n$$

Commulative generating fn

$$K_x(t) = n \log \left(\sum_{i=1}^K p_i e^{t_i} \right)$$

hypergeometric Distribution.

$$P(X=x) = \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}$$

$$\mu = np$$

$$\sigma^2 = npq \left(\frac{N-n}{n-1} \right)$$

$$p = \frac{K}{N}, \quad q = \frac{N-K}{N}$$

Poisson Distribution.

$$P(x) = \begin{cases} \frac{e^{-\lambda} \lambda^x}{x!} \\ 0 \end{cases}$$

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$$\mu = np$$

$$\sigma^2 = npq$$

Negative Binomial Distribution

$$P(X=x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}$$

$$\text{mean} = E(x) = \frac{r(1-p)}{p}$$

$$\text{var}(x) = \frac{r(1-p)}{p^2}$$

$$m(t) = \left(\frac{1-p}{1-pe^t} \right)^r \quad t < -\log p$$

Geometric Distribution

$$P_x(x) = (1-p)^{x-1} p$$

$$\mu = \frac{1}{p}$$

$$\sigma^2 = \text{Var}(x) = \frac{1-p}{p^2}$$

pmf of multinomial Distⁿ

$$\left\{ \frac{n!}{x_1! x_2! \dots x_k!} p_1^{x_1} p_2^{x_2} \dots p_k^{x_k} \right.$$

mean

$$E(x_i) = np_i$$

$$\text{Var}(x_i) = np_i(1-p_i)$$

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Discrete uniform distribution

$$\text{var}(x) = E(x^2) - (E(x))^2$$

$$\sigma = \sqrt{\frac{n^2 - 1}{12}}$$

$$\text{mean} = \mu = \frac{n+1}{2}$$

moment generating fn.

$$m(t) = \frac{1}{n} \sum_{x=1}^n (e^t)^x$$

Sum of geometric series

$$= \frac{a(1-r^n)}{1-r}$$

Binomial Distribution

$$P(x) = \binom{n}{c} p^c (1-p)^{n-c}$$

$$\text{mean } \mu = np$$

$$\text{var}(x) = np(1-p)$$

$$\sigma = \sqrt{np(1-p)}$$

$$m(t) = \left[(1-p) + pe^t \right]^n$$

$$\text{mean} = \mu = m'(0)$$

$$\sigma^2 = m''(0) - \mu^2$$